Data Structures Made Easy with Java Collections

This article introduces the reader to Java Collections, which is a framework that provides an architecture for collecting and manipulating a group of objects. Operations such as searching, sorting, insertion, manipulation, deletion, etc, can easily be performed by Java Collections.

The modern IT world revolves around data. Every new site asks for a sign-up and, in the process, collects information and saves it. There is cut-throat competition in gathering data. Sites that have a large user base and hence a vast store of data have the upper hand. Whether it is Google, Amazon, Flipkart, WhatsApp, Facebook or Twitter, all of them gather information and save it because data is of utmost importance in the present day world.

Data in its raw form is like a chunk of ore from which the diamond is yet to be extracted. It is hard to process the raw data so the need for structured data arises, which is when data structures become handy. In computer science, a data structure is a particular way of organising data in a computer so that it can be used efficiently. Searching a particular book from millions of books on Amazon or sorting through the endless choice of mobile phones on the basis of price are all done with low cost and low complexity algorithms, which work on structured data.

Data can be structured in many ways but the commonly used practices are given below: